Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 310 :: Software Development Project-I

Project Report

Project Title : Medicore

Submission Date : Mon 01, Dec 2025

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Medicore

1. Introduction

Medicore is a cross-platform mobile application developed to streamline and modernize healthcare services. It integrates digital appointment booking, AI-based symptom analysis, a medicine store, and secure payment processing into a single platform. Designed for patients, doctors, and administrators, Medicore aims to bridge gaps in traditional healthcare systems by offering a unified, efficient, and user-friendly digital solution.

2. Objectives

2.1 Primary Goals To develop a scalable and intuitive medical service platform using Flutter. To implement real-time backend services with Firebase (Authentication, Firestore, Storage). To integrate Google Gemini AI for preliminary symptom analysis and health recommendations.

2.2 Secondary Goals To provide role-based dashboards for patients, doctors, and admins. To enable secure online payments via SSLCommerz. To ensure robust state management using the Provider pattern.

3. Problem Statement

Traditional healthcare systems face challenges such as: Manual and time-consuming appointment processes. Lack of integrated digital health records. Limited access to preliminary medical guidance. Inefficient communication between patients and healthcare providers. Medicore addresses these issues by offering a centralized platform for appointments, AI-driven health insights, e-commerce (medicines), and secure data handling.

4. Related Work

Existing medical apps like Practo and Halodoc offer similar services but often lack: AI-powered preliminary diagnosis. Localized payment gateways (e.g., SSLCommerz). Integrated medicine delivery within the same app. Medicore differentiates itself through its AI-driven symptom checker, multi-role support, and localized payment integration.

5. Scope

Development of a cross-platform app (Android, iOS, Web) using Flutter. Integration of Firebase for authentication, database, and storage. Implementation of AI-based health suggestions via Gemini API. Secure payment processing with SSLCommerz. Role-based access for patients, doctors, and admins.

6. Methodology

6.1 Technology Stack Component Technology/Service Frontend Flutter (Dart) Backend Firebase (Auth, Firestore) State Management Provider AI Engine Google Gemini API Payment Gateway SSLCommerz Cloud Storage Cloudinary

6.2 Design Principles Material Design: Consistent and modern UI across platforms.

Role-Based Access Control: Separate interfaces for patients, doctors, and admins.

Security: Secure authentication, data encryption, and PCI-compliant payments.

Scalability: Modular codebase for future enhancements.

7. Visual Models

7.1. Flow Chart Diagram

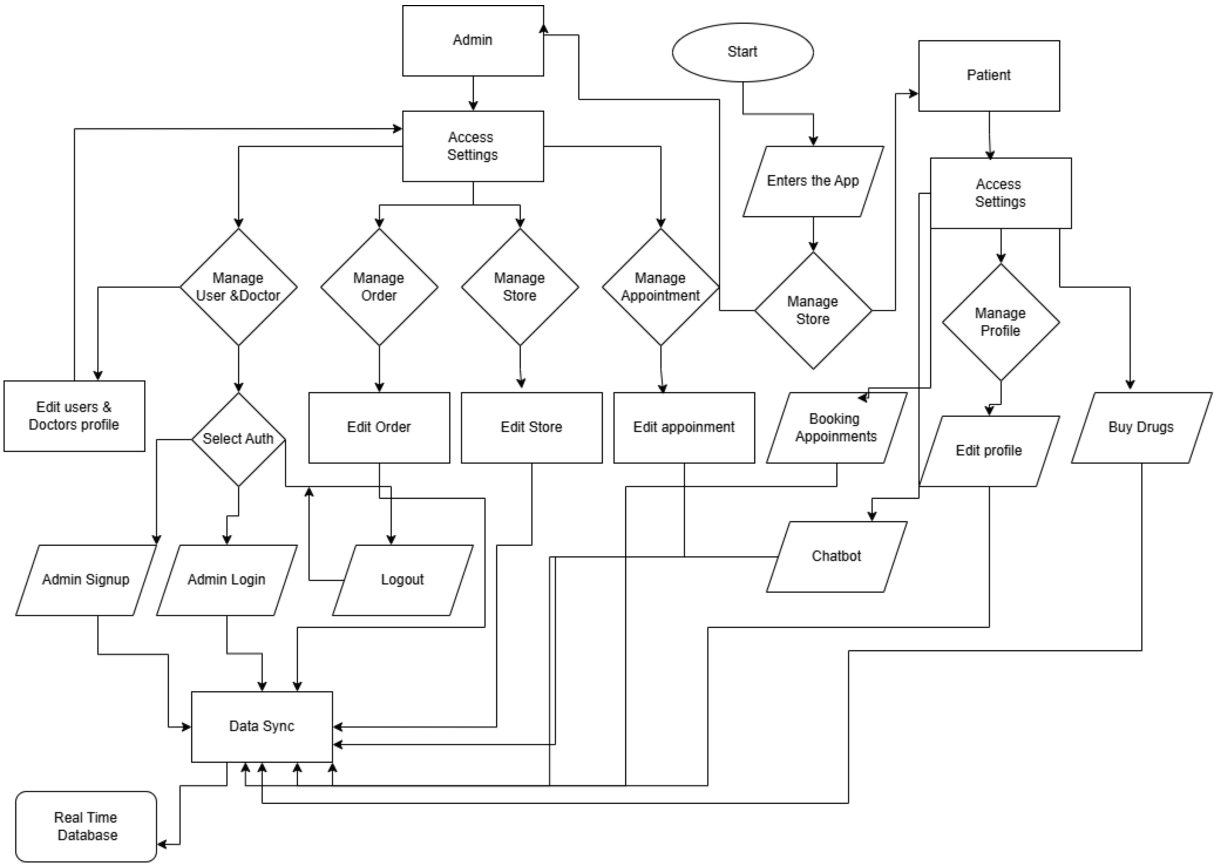

Figure 1: Flow Chart of Medicore Architecture

The above Figure 1 illustrates the overall architecture of the Medicore application, showing the interaction between users, the frontend application, backend services, and the database. It highlights the flow of data and the key components involved in the system.

7.2. Schema Diagram

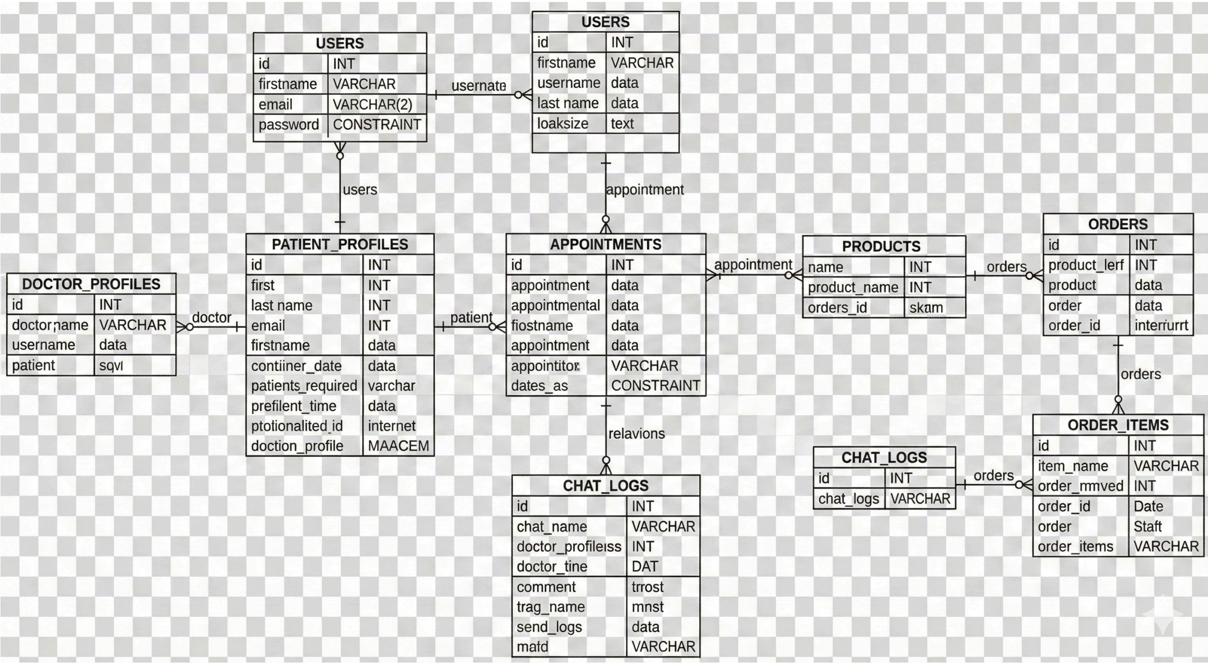

Figure 2: Database Schema Diagram of Medicore

The above Figure 2 illustrates the database schema for Medicore, showing the tables, their fields, and relationships between them. The schema is designed to efficiently store and retrieve contact information and user data.

7.3. ERD (Entry Relationship Diagram)

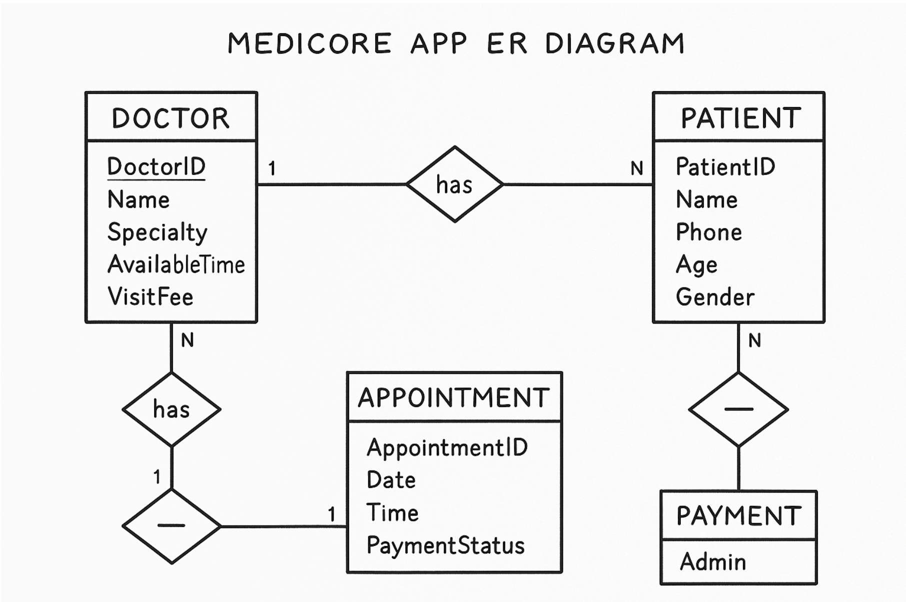

Figure 3: Entity Relationship Diagram of Medicore

The above Figure 3 illustrates the entity-relationship diagram for Medicore, showing the entities, their attributes, and relationships between them. The ERD helps in understanding the data model and how different entities interact with each other.

7.4. Timeline (Gantt Chart)

The base timeline for the development of Medicore is as follows,

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9	Week 10	Week 11	Week 12
Requirements & UI Mockup	✓		✓					
App structure and navigation design		✓	✓					
Basic UI and supabase/ backend integration			✓	✓				
Admin panel design with role-based access control			✓	✓	✓			
Fetch data realtime from db						✓		
UI Polish & Documentation						✓	✓	
Final Testing & Deployment								✓

Table 1: Development Timeline of Medicore

The timeline is divided into 12 weeks, with specific tasks allocated to each period which describes an approximate timeline for the whole development process.

7.5. UI Mockups

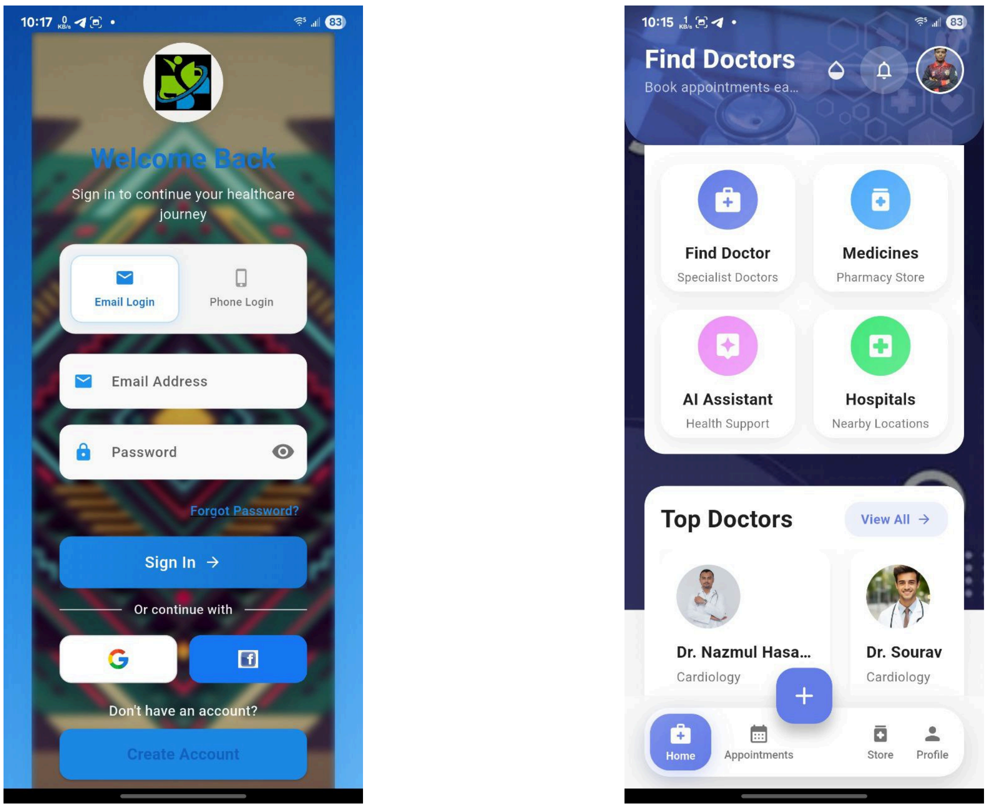

Figure 4: Add Bookmark page and reader page's UI concept

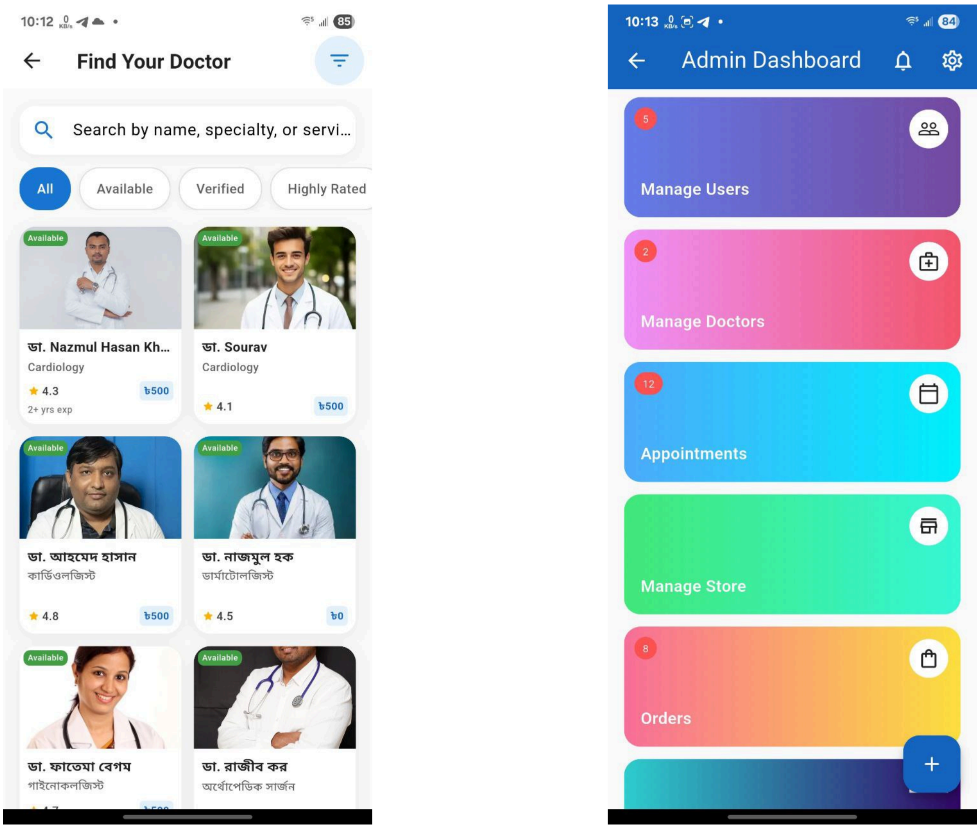

Figure 5: Settings and sign in page's UI concept

8. Future Plans

1. Telemedicine: Integration of video consultations via WebRTC.
2. Electronic Health Records (EHR): Digital prescriptions and patient history.
3. Wearable Integration: Sync with fitness trackers (Google Fit, Apple HealthKit).
4. Multi-Language Support: Expand accessibility for non-English speakers.
5. Advanced Analytics: Predictive health insights using machine learning.

9. Result

Medicore successfully delivers a unified platform for digital healthcare management. Key achievements include:

- Seamless appointment booking and payment processing.
- AI-driven symptom analysis for preliminary guidance.
- Secure and scalable architecture with Firebase.
- Enhanced user experience for patients, doctors, and administrators.

The app is poised to improve healthcare accessibility, reduce administrative overhead, and provide intelligent health insights.

10. References

1. Flutter Documentation: https://flutter.dev
2. Firebase Documentation: https://firebase.google.com
3. SSLCommerz Documentation: https://sslcommerz.com